# A bean becomes.

## 01 What is waking up inside this little bean?



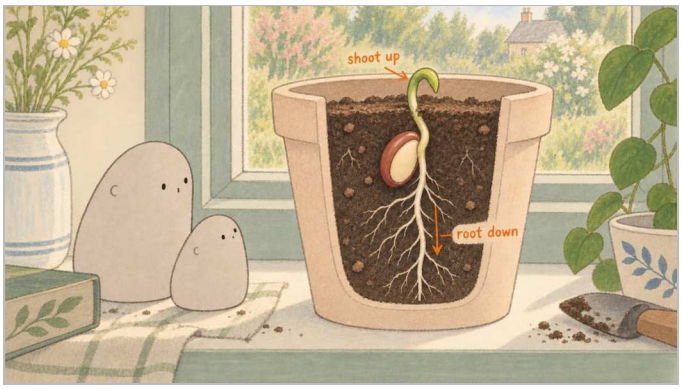

An imagined window into the soil. The seed is enlarged to help us see.

**Look closer.** Look inside the seed coat. What might those two pale halves be keeping safe?

**A little explanation.** A tiny plant is already inside. The bean holds food to help it begin.

**Try this.** Curl one hand like a seed. Slowly uncurl a finger for its first root.

## 02 How does a root find its way when it cannot see?




Roots and shoots respond to their surroundings. Plants do not need eyes.

**Look closer.** Follow the white root down. Now follow the little shoot up. What do you notice?

**A little explanation.** The root does not need eyes. It responds to gravity, which helps it grow down.

**Try this.** Follow the two directions in the air: root down, shoot up.

## 03 What can a leaf do that a seed cannot?




Leaves make new food using light, water, and carbon dioxide from the air.

**Look closer.** Find the three notes around the leaves. Where does each arrow lead?

**A little explanation.** Green leaves use light, water, and part of the air to make new food. The seed began with food stored inside.

**Try this.** Follow a path from the roots to a leaf. Then point to the light.

## 04 How could we notice a change too slow to watch?




Compare from the soil level. This painted stick is not a scale ruler.

**Look closer.** Find the earlier mark and today's tip. What could a drawing help us remember?

**A little explanation.** We can compare notes from different days. Growth can happen too slowly for our eyes to catch it happening.

**Try this.** Hold a finger at the earlier mark. With your other hand, show the space up to today's tip.